



ACADEMIC PROJECT REPORT

Title: Management Consulting Analytics in ESG and Financial Performance

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Introduction

In today's corporate landscape, Environment, Social and Governance (ESG) performance is no longer a symbolic gesture, it is a strategic determinant of firm value; investment attractiveness, and long term sustainability. From institutional investors to regulators, stakeholders increasingly demand not only ESG disclosures but evidence that ESG actions create measurable financial impact

While large corporations in developed markets have begun aligning ESG strategies with financial outcomes, mid-sized firms in economies like Indonesia remain at a critical disadvantage. These firms face mounting pressure to improve ESG performance but lack clear, data backed insight into how ESG metrics influence probability, valuation or financial stability.

Existing research overwhelmingly focuses on macro trends, sector agnostic datasets, or developed-market scenarios. This creates a practical gap: firms do not know which ESG initiatives drive returns and managers are often relying on perception over performance.

This project addresses that gap through application-based, analytics driven framework. It explores whether stronger ESG performance is associated with better financial outcomes, specifically – Return on Assets (ROA), Tobin's Q, Sustainable growth rate (SGR) in Indonesian publicly listed companies. By engineering new diagnostics indicators such as –

- 1) Growth adjusted ESG
- 2) Profitability gap
- 3) Leverage-to-ESG ratio

And visualizing ESG financial linkages through excel models and tableau dashboards, the project provides decision-useful tools for consulting professionals, CFOs, ESG officers and investors

The study moves beyond descriptive ESG reporting into diagnostic performance analytics, aligning academic inquiry with real world consulting needs. It ultimately aims to help organisations prioritize ESG investments that generate financial value, rather than symbolic compliance.

Literature Review: Management Consulting Analytics in ESG and Financial Performance

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1. Introduction

In recent years, Environmental, Social, and Governance (ESG) performance has transitioned from a symbolic indicator of corporate responsibility to a critical component influencing firm valuation, probability, and risk. While numerous studies confirm ESG's potential alignment on financial metrics, gaps remain regarding its practical application, especially in emerging markets like Indonesia. This literature review synthesizes over 50 scholarly articles to critically analyse how ESG performance correlates with financial performance, emphasizing analytics-driven insights for consulting applications. It integrates this understanding with an applied methodology involving regression diagnostics, profitability gap analysis, and sectoral benchmarking.

2. Global Evidence on ESG–Financial Performance Link

(Gunnar Friede, 2015) looked at more than 2000 studies and found that about 90% showed no downside between ESG and financial performance. So, most of the time, ESG doesn't hurt returns. (Upadhyay, 2025) went further and showed that ESG-focused companies actually beat their peers when it comes to stock returns. (Joshua D. Margolis, 2011) conducted a meta-analysis affirming that CSR and ESG initiatives generally produce a modest but positive effect on financial returns

(Paolo Fiorillo) found that companies with higher ESG scores pay less to borrow money. In simple terms, being good at ESG helps with creditworthiness.

(Rui A. Albuquerque, 2018) saw that strong sustainability practices lead to less risk and lower volatility for firms. But not everything is straightforward. (Florian Berg, 2022) pointed out that ESG ratings aren't consistent across providers, which makes it tough to compare companies or make solid decisions. On the other hand, (Luis Jacob Escobar-Saldívar, 2025) found that companies improving their ESG scores see less volatility in U.S. markets and so both where a company stands and how it's moving matters. (Smita Dayal, 2024) further explored how ESG disclosure quality influences both firm risk and operational performance across global markets, underscoring the disclosure-performance linkage.

3. ESG in Emerging Markets: Evidence and Gaps

There's plenty of research on how ESG ties into financial performance in developed countries, but things get more interesting in developing economies. (Shamsul Nahar Abdullah, 2022) and (María Cantero-Saiz S. S.-A.-O., 2025) point out that the story changes depending on the local context. For example, (María Cantero-Saiz S. S.-A.-O., 2025) saw ESG helping companies in Pakistan become more competitive and profitable.

Then there's, (Tripopsakul, 2025), who looked at firms in ASEAN countries. He found that when companies back green innovation with solid ESG policies, their financial performance gets a real boost. (Mercedes Daimler Benz Ernanda, 2025) focused on Indonesia's energy sector and confirmed that ESG matters for firm value and even when the regulations are all over the place. (Egi Arvian Firmansyah, 2023) examined Saudi-listed companies and found that ESG disclosure

significantly improved ROA and firm reputation, though the effect on market value remained mixed.

Still, a lot of the research in emerging markets sticks to big picture trends and doesn't really dig into specific sectors. Plus, many studies use broad, country-level data instead of looking closely at individual companies. That's why this project zeroes in on firm-level, industry-specific analysis and exactly what's missing from most of the work out there. (Hariom Manchiraju, 2018) studying India's CSR mandate, found that such regulatory enforcement improved firm valuation only when paired with strategic intent.

4. Sectoral Variation and ESG Sensitivity

(Atif Mahmood, 2024) demonstrated that ESG's alignment on firm value varies significantly across sectors, particularly in EU-listed firms. Similarly, (Auliya Zahra, 2019) emphasized the heterogeneous effects of sustainability practices. (Peliu, 2024) identified a stronger ESG financial performance link in highly regulated industries. (Elisa Menicucci, 2024) found that governance factors (e.g., women on boards) significantly drive ESG performance in the utilities sector. (Ozgur Ozdemir, 2021) found that gender-diverse boards significantly influence the strength of ESG implementation, particularly in service and hospitality industries. (Chien-Ming Wang, 2025) emphasized non-linear ESG-performance dynamics in tourism, suggesting that returns to ESG investment plateau after a certain threshold.

This supports our project's emphasis on sector-specific analytics, using ROA, Tobin's Q, and ESG-normalized scores across Indonesian sectors such as finance, materials, and consumer discretionary.

5. Methodological Approaches in Prior Literature

Quantitative methodologies dominate ESG research. (Hua Nian, 2024) provided a systematic review, highlighting the use of regression models, panel data analysis, and event studies. (Tensie Whelan, 2020) aggregated over 1,000 studies to develop a decision-useful ESG-financial framework. (Ícaro Guilherme Félix Cunha) proposed a conceptual framework integrating ESG metrics with corporate performance indicators, advocating for clearer standardization in measurement.

(Mozaffar Khan, 2015) emphasized the importance of materiality mapping, while (Zhuo Li, 2025) focused on investment efficiency metrics. (Peter Clarkson, 2007) tested whether it pays to be green using industry-adjusted metrics. (Diane-Laure Arjaliès, 2015) examined how management control systems can be aligned with CSR strategy using a levers-of-control approach.

These methods align with my project approach, which includes : ESG quartile segmentation, profitability gap analysis (vs. sector averages), leverage-to-ESG ratios, and regression modeling.

6. ESG, Risk, and Value Creation

(Witold Henisz) proposed five mechanisms through which ESG creates value, including better resource optimization and regulatory alignment. (Sanchez, 2025) Found that sustainability officers significantly influence firm performance.

(Petit, 2019) demonstrated that ESG news influences stock price volatility. (Rio Murata, 2021) linked ESG disclosure to lower crash risk, reinforcing the relevance of transparency. (FOUQUAU) explored investor attention dynamics, showing how ESG-related news triggers significant shifts in financial market sentiment.

Our consulting-style project mirrors these insights through a diagnostic lens – visualizing firm performance relative to ESG quartile, stress-testing financial stability through leverage ratios, and mapping value gaps. (Amir Amel-Zadeh, 2020) Surveyed global investors and confirmed ESG factors are increasingly embedded in investment decision-making, particularly for long-horizon portfolios.

7. Rating Systems and Reporting Challenges

One thing that keeps coming up in the research is just how inconsistent ESG ratings are. (Florian Berg, 2022) actually called it “aggregate confusion”. (Robert G. Eccles L.-E. L., 2020) pointed out that most of this mess comes from rating agencies using different methods. (Philipp Krueger, 2019) stressed the financial materiality of climate risk disclosures, adding urgency to harmonize ESG frameworks. (Robert G. Eccles I. I., 2012) made a connection between a company’s sustainability culture and how well it does over the long haul, but they also warned that its tough to compare companies when the metrics keep shifting.

This project works around those problems. It sets up a standardized ESG framework and factors in differences between sectors, so the analytics actually help when it comes to making consulting recommendations. (Boffo, 2020) identified persistent challenges in ESG investing such as inconsistent taxonomies, which complicate performance attribution across funds.

8. Consulting Implications and Strategic Insights

(Witold Henisz) and (Diane-Laure Arjaliès, 2015) looked at how companies can actually build ESG into their management control systems. (Mohammed Benlemlih, 2018) showed that CSR doesn’t just look good on paper but it really helps companies invest smarter, which backs up the idea that ESG has real strategic value. (Steve Lydenberg, 2009) discussed the strategic future of responsible investing, calling for ESG models that are both value-aligned and evidence-based.

With our project, we take these academic ideas and turn them into tools people can actually use. We’ve built visual dashboards in Tableau, mapped ESG quartiles so you can see how companies stack up against each other, run regression diagnostics to predict profitability, and created prioritization matrices to guide where ESG capital should go.

(Xiaoling Yu, 2022) introduced a new ESG-scoring model tailored for Chinese enterprises, offering greater predictive accuracy of firm value. (Mozaffar Khan, 2015) demonstrated that only material ESG issues – those tied to sector-specific risks- correlate strongly with superior financial outcomes. All of this lines up with what you’d find in real consulting work. It’s grounded in research, sure, but its also practical and ready to use in the real world.

9. Conclusion:

This project bridges a critical gap between academic ESG research and consulting analytics, focusing on the financial relevance on ESG performance in Indonesian listed companies. While global literature broadly supports a positive relationship between ESG and financial outcomes, most studies overlook firm level variation, sectoral nuances, and decision useful modelling especially in emerging markets.

By adopting a diagnostic approach, this study moves beyond theory into actionable insight. Using metrics like ESG-normalised scores, profitability gap analysis, Tobin’s Q and Leverage-to-Esg ratio, the project uncovers patterns that helps firms understand that where ESG aligns with financial

value. Regression modelling and sectoral benchmarking further reveals that ESG alignment is non uniform, its highly context-dependent.

The project contributes both practically and academically by:

- Translating complex ESG frameworks into firm-level, consultable KPI's
- Visualising ESG-financial linkages through tableau dashboards
- Building prioritization tools that can guide ESG capital allocation decisions.

Ultimately, this work serves as a foundation for evidence-based ESG strategy in consulting, empowering firms to treat ESG not just as a reporting obligation, but as a lever for competitive financial performance.

10.Limitations

The analysis focuses solely on Indonesian publicly listed companies. While this strengthens local relevance, it limits generalizability to other markets, particularly those with different regulatory and ESG reporting environment. This study takes cross-sectional approach using the most recent data. The methodology focuses on statistical relationships (regression, correlation) rather than casual inference.

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Problem Statement

Potential Contribution to existing Knowledge or Novelty or Innovation in approach

ESG frameworks are changing the way companies talk about more than just profits. They are everywhere now but let's be real – most companies, especially in developing markets still have no clue how to turn those ESG scores into something that actually matters for their bottom line. Everyone sees the rating, but what do they really mean for a company's financial health? The link is fuzzy and it depends a lot on the industry. Most people aren't just measuring it right.

This research takes on that problem head-on. Instead of talking about ESG in the abstract, it lays out a new consulting approach built around analytics. The big question : Do companies with better ESG performance actually make more money? We are talking about real numbers here – Returns on Asset(ROA), Tobins Q, Sustainability growth rate. To dig deeper, the study introduces some fresh tools at the firm level:

1)Profitability Gap

2)Growth adjusted ESG Scores

3)Leverage-to-ESG ratio

These aren't your average, catch all ESG metrics. The researcher designed them to give companies a sharper, more specific look at their own strengths and weaknesses – tailored by sector not just broad averages. By mixing statistical models with data visualization in Tableau, the project doesn't just crunch the numbers. It also makes the insights easy to see and act on. That's what sets this work apart and it's why it matters for consultants, investor relations teams, and anyone trying to get real about ESG strategy.

Contemporary Relevance of the research

ESG reporting isn't just a nice-to-have anymore – lots of places now require it and for mid-sized companies, that means they have to prove their ESG spending pays off, not just in good vibes, but in real numbers. The problem? Data's messy, ratings don't lineup and there's no clear roadmap. So for many, ESG still feels like just another box to tick instead of something that actually boosts performance.

This project hits right where it matters, especially after 2020. Investors, regulators, and company boards all want proof that firms are serious about ESG. Looking at Indonesian public companies makes things even more interesting. That markets growing fast, but people haven't really dug into its ESG efforts yet. And the timings couldn't be better – with everyone from stakeholders to international groups (like ISSB, GRI and SASB) putting for more consistent sustainability, reporting the pressure is on.

Objectives

- 1) Measure how ESG performance actually links to financial results – things like ROA, Tobin's Q, and Sustainability Growth Rate. I'm using regression and correlation analysis to nail down those connections.
- 2) Spot how these ESG-financial relationships play out in different sectors. I break things up by quartile, run industry benchmarks, and look at profitability gaps across categories to see who's leading and who's lagging.
- 3) Create some sharper ESG-financial metrics like Growth-adjusted ESG, Leverage-to-ESG Ratio, and a Profitability gap score. The goal? Make it way easier to see how much value ESG really brings.
- 4) Build an ESG impact index and visual dashboards by Tableau and Excel so companies can actually track their ESG, ROI and Zero in on the sustainability moves that pay off the most.
- 5) Sort companies into groups: high-ESG high-return versus high-ESG low-return. This way firms can make better calls about where to put their ESG investments, based on real world diagnostic data.

Scope of Research

This study focuses on evaluating the financial implications of ESG performance Indonesian publicly listed firms with a concentrated lens on the financial sectors basic materials and consumer discretionary sectors. These sectors were selected for their varied ESG maturity levels and financial structures, making them ideal for benchmarking ESG-financial alignment.

The scope is sharply defined across four operational dimensions:

1) **Geographic market scope:**

The study is limited to Indonesia's capital market, representing firms, reflecting emerging market ESG challenges.

The analysis targets mid-to-large firms with publicly disclosed ESG and financial data.

2) **Analytical Scope:**

Focuses on evaluating cross-sectional ESG-financial performance linkages, not longitudinal trends.

Includes financial variables like-ROA, Tobin's Q, Sustainable Growth Rate, and Leverage-to-ESG Ratio, benchmarked against engineered ESG metrics.

3) **Data Scope:**

Uses secondary data extracted from sustainability reports, annual financial disclosures and ESG score database.

Introduces derived metrics : Profitability gap, Growth adjusted ESG, Leverage Ratio for diagnostic analysis.

4) **Application Scope:**

Designed to stimulate a consulting project, the research goes beyond academic interpretation to build Tableau dashboards, prioritization tools, and regression based diagnostics to help firms:

- Evaluate ESG maturity
- Identify high-ESG / high return combinations
- Align ESG strategy with profitability targets

The study does not attempt to measure causality or account for policy or regulatory impact in depth. Instead it aims, to provide sector-specific, decision useful insights for managers, investors and ESG analysts operating in emerging markets.

Research Questions

This study seeks to answer the following seven research questions using a diagnostic, analytics driven approach to ESG – financial performance evaluation:

- 1) What is the relationship between ESG performance and financial outcomes such as ROA, Tobins Q, and sustainable growth rate in Indonesian listed firms?
- 2) Do firms in the top ESG quartile consistently outperform those in lower quartiles in terms of profitability and market valuation?
- 3) How does ESG performance interact with firm leverage to influence financial risk and capital efficiency?
- 4) Which sectors demonstrate the highest return per unit of ESG score, and how does this vary across financial, consumer and basic materials industries?
- 5) What insights can be drawn by comparing firm level profitability gaps relative to sector averages across ESG quartiles?
- 6) Can newly engineered metrics like Growth-adjusted ESG and Leverage-to-ESG ratio improve ESG impact assessment over conventional ESG scores?
- 7) How can analytics tools such as regression models and Tableau dashboards be used to develop a standardized ESG impact Index for consulting use?

Hypothesis

Purposes of Hypothesis Formulation

The hypothesis provides a statistically testable foundation to examine the link between Environmental, Social and Governance (ESG) performance and financial outcomes of firms. Since the research design is diagnostic and correlation-based, hypothesis are framed to test directional relationships using regression analysis and sectoral comparison.

Independent Variables (Predictors)

- 1) ESG Performance Score (raw ESG metrics)
- 2) ESG Normalized Score (scaled 0-100)
- 3) ESG Quartile Classification (categorical : Top, High, Medium and Low)

Dependent Variables (Financial Outcomes)

- 1) Return on Assets (Proxy for profitability)
- 2) Tobin's Q (Proxy for market valuation)
- 3) Sustainable Growth Rate (Proxy for long term value creation)

Control/Diagnostic Variable

- 1) Leverage Ratio (financial risk)
- 2) Corporate sector (Industry Control)
- 3) Profitability Gap (Relative Sector Performance)
- 4) Growth-Adjusted ESG (Engineered indicator)
- 5) Leverage-to-ESG Ratio (Risk-efficiency metric)

Null Hypothesis (H0)

- H01 : ESG Performance does not significantly effect ROA of Indonesian listed firms.
- H02 : ESG performance does not significantly effect Tobin's Q of firms.
- H03 : ESG performance has no significant relationship with significant growth rate
- H04 : There is no significant difference in financial performance across ESG Quartiles.
- H05 : ESG-Financial impact does not vary significantly across sectors.

Alternate Hypothesis (H1)

- H11 : Higher ESG performance is positively associated with ROA.
- H12 : ESG performance is positively associated with Tobin's Q.
- H13 : ESG performance positively predicts Sustainable Growth Rate.

- H14 : Financial performance varies significantly across ESG Quartiles.
- H15 : ESG-Financial performance linkage is moderated by sector classification.

Summary

These hypothesis ensure alignment between the objectives, diagnostic indicators and statistical testing approach (Excel + Tableau). Each hypothesis is built to test measurable ESG – Financial linkages, enhancing both academic validity and consulting relevance.

Research Methodology

This study adopts a quantitative, application-based and diagnostic research methodology to examine the relationship between ESG performance and financial performance metrics of Indonesian publicly listed companies. The approach is designed to reflect real world consulting frameworks and meets the criteria for applied, industry – relevant academic research.

Research Approach and Types

The study follows a correlational and diagnostic analytical design, supported by:

- Descriptive Analysis – for ESG and financial performance trends
- Inferential Analysis – to test relationships using correlation and regression
- Benchmarking – to compare performance across sectors and ESG quartiles.
- Visualization – via Tableau dashboards for pattern recognition and communication.

Data Source and Tools Used

The study used secondary data and sourced from :

- ESG performance datasets
- Financial statements and sustainability disclosure of listed companies

Tools and platforms:

- Microsoft Excel – for data preparation, computation, pivot (tables and charts) and regression analysis.
- Tableau – for interactive visualization and dashboarding
- MSWord and PowerPoint – for documentation and presentation and insights

Variables and constructs

Independent variables (ESG):

- ESG Raw Score
- ESG Normalized Score

- ESG Quartile classification

Dependent Variables (Financial):

- Returns on Asset
- Tobin's Q
- Sustainable Growth Rate

Diagnostic Variable:

- Sector classification
- Leverage Ratio
- Profitability Gap
- Growth-Adjusted ESG
- Leverage-to-ESG Ratio

Data Processing and Feature Engineering

Feature engineering was performed in Excel to create:

- Profitability Gap = $ROA(\text{firm}) - ROA(\text{sector average})$
- Growth-Adjusted ESG = $ESG \text{ Score} * \text{Sustainable Growth Rate}$
- Leverage-to-ESG Ratio = $\text{Leverage} / ESG \text{ Score}$

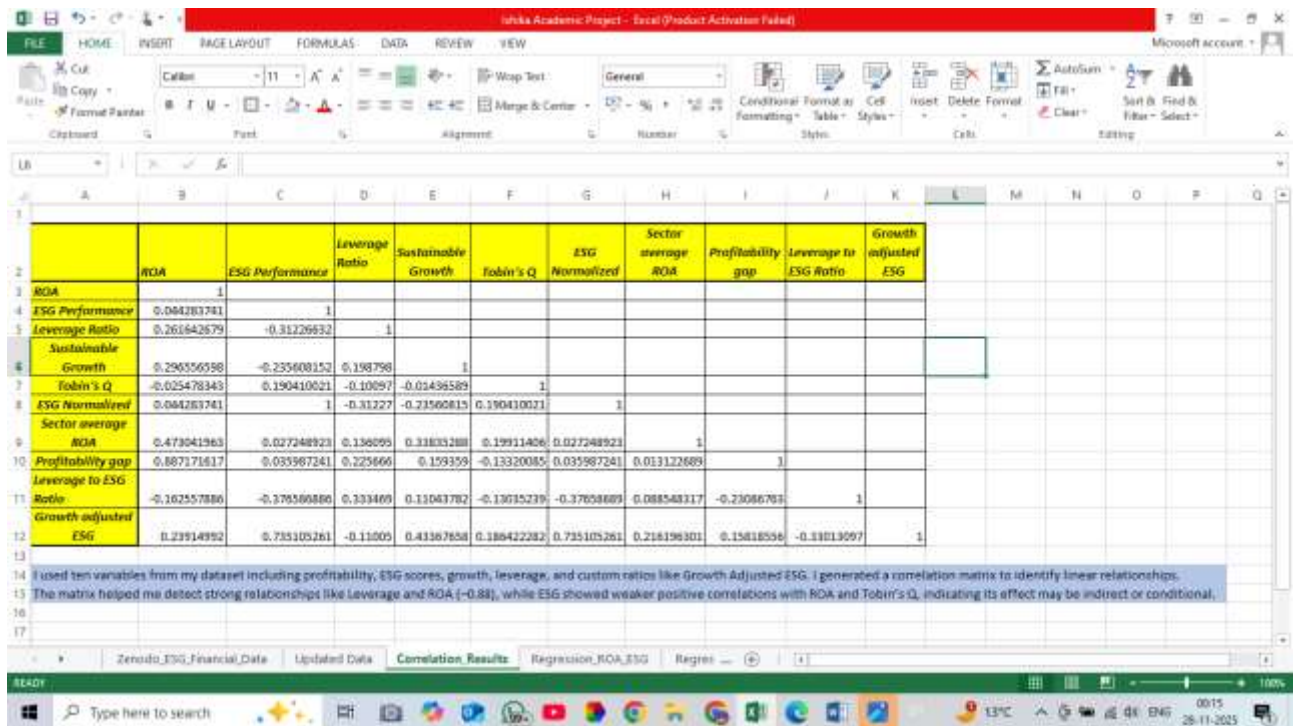
Year	Code	Company's Name	Corporate Sector	RoA	ESG Performance	Leverage Ratio	Sustainable Growth	Tobin's Q	ESG Normalized	Sector Average ROA	Profitability Gap	Leverage to ESG Ratio	Growth Adjusted ESG	ESG Quartile
2023	ANTM	Aneka Tambang Tbk	Basic-Materials	0.05	34.72	0.36	0.14	0.1	77.03408044	0.15	-0.1	0.094673254	10.78477128	Top
2023	AUTO	Astra Otoparts Tbk	Consumer Discretionary	0.11	28.14	0	0.16	0.07	56.83143998	0.126666667	-0.016666667	0	9.093030396	High
2023	BBCA	PT Bank Central Asia Tbk	Financials	0.08	21.26	0.15	0.49	0.03	35.52348787	0.064333333	-0.034333333	0.004222558	17.40650906	Moderate
2023	BBNI	PT Bank Negara Indonesia (Persero) Tbk	Financials	0.02	22.90	0.12	0.34	0.02	40.74301564	0.064333333	-0.044333333	0.00294529	13.85262512	Moderate
2023	BBRI	PT Bank Rakyat Indonesia (Persero) Tbk	Financials	0.05	17.70	0.15	0.28	0.06	34.77740252	0.064333333	-0.034333333	0.006053903	6.937672705	Low

These indicators offer granular diagnostics to identify high score – low return or high ESG – high returns.

Statistical Tests Applied

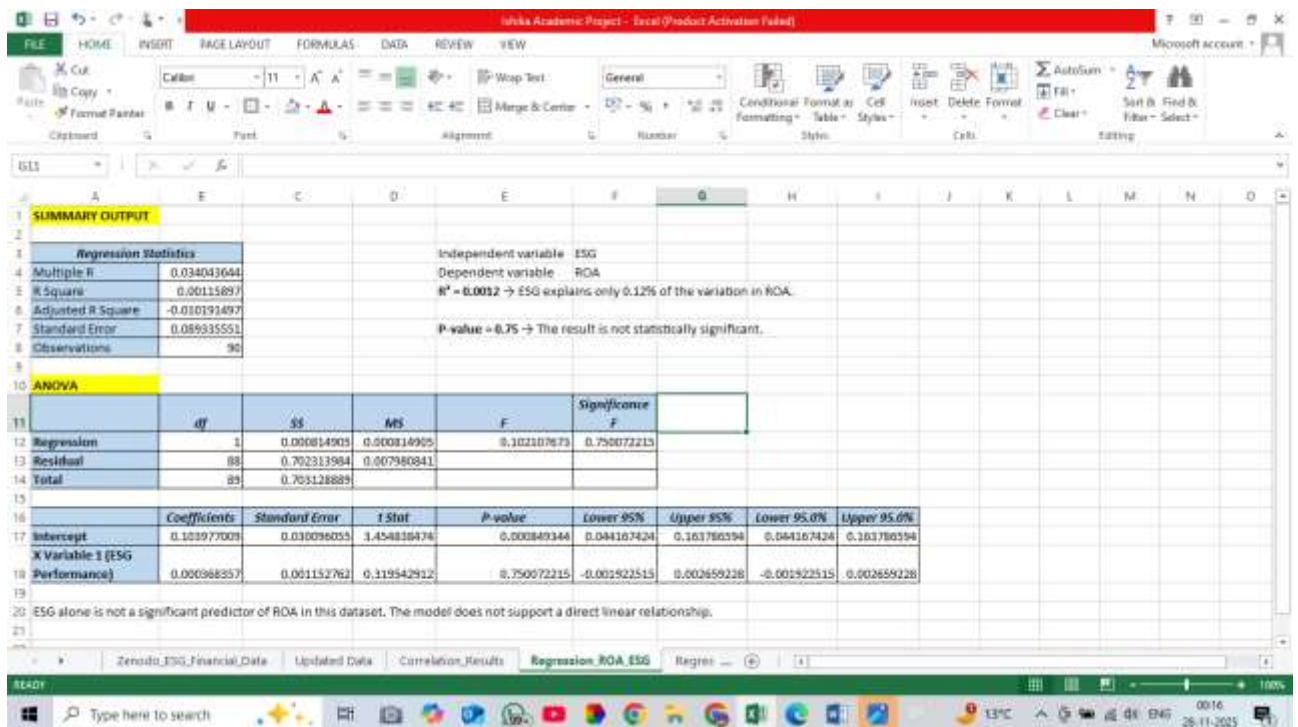
Correlation Analytics:

- To assess initial relationships between ESG Scores and financial variables



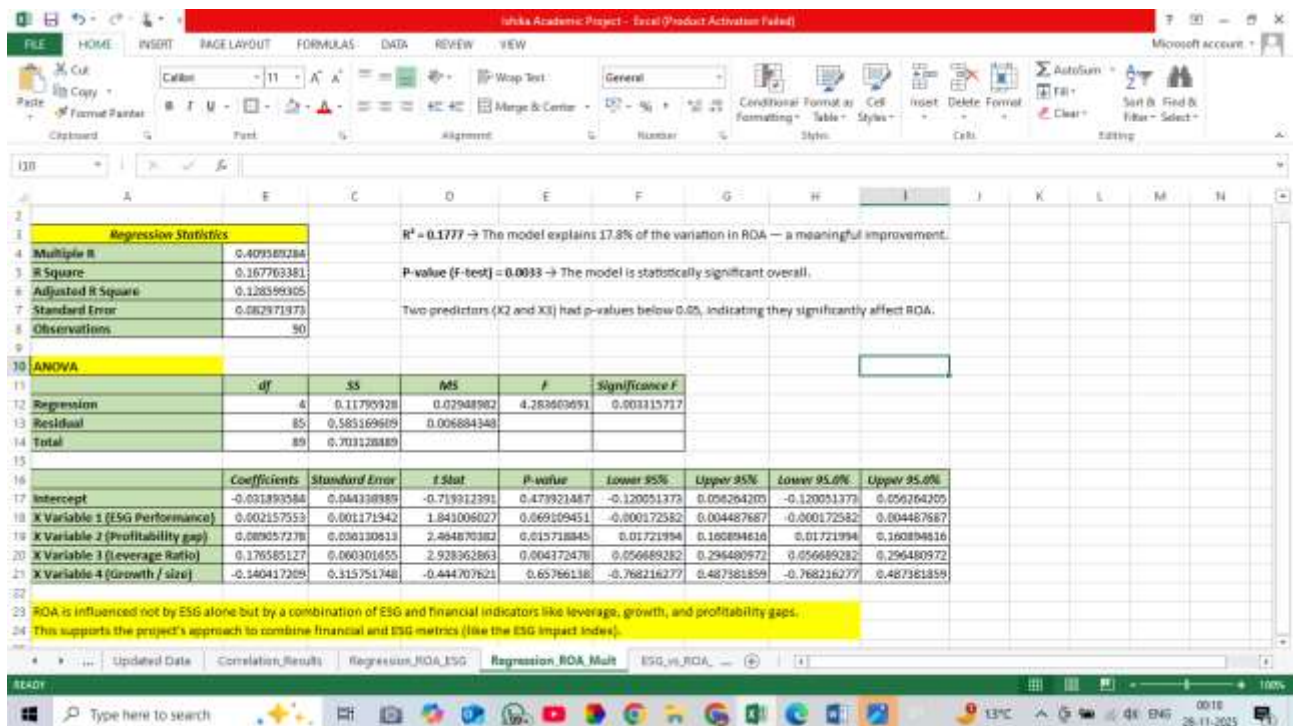
Simple Regression Model:

- $ROA = \beta_0 + \beta_1(\text{Financial Performance})$



Multiple Regression Model:

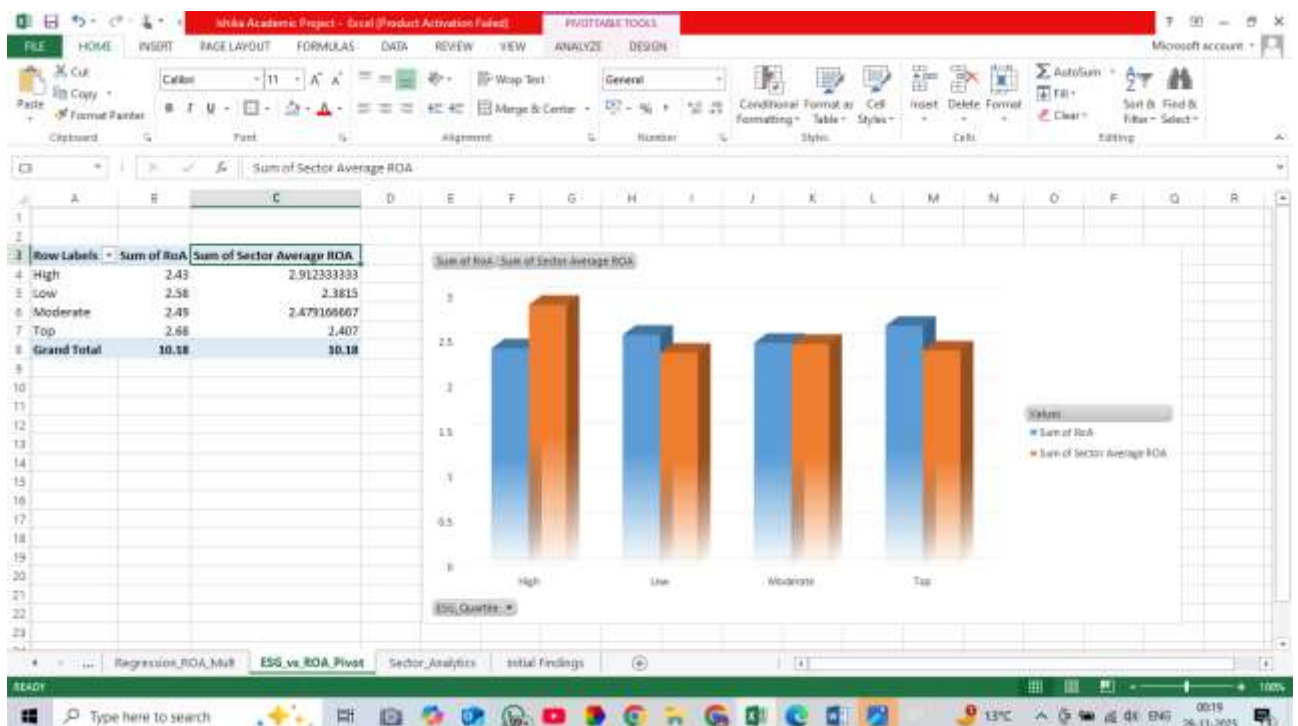
- $$ROA = \beta_0 + \beta_1(\text{ESG Performance}) + \beta_2(\text{Leverage ratio}) + \beta_3(\text{SGR})$$



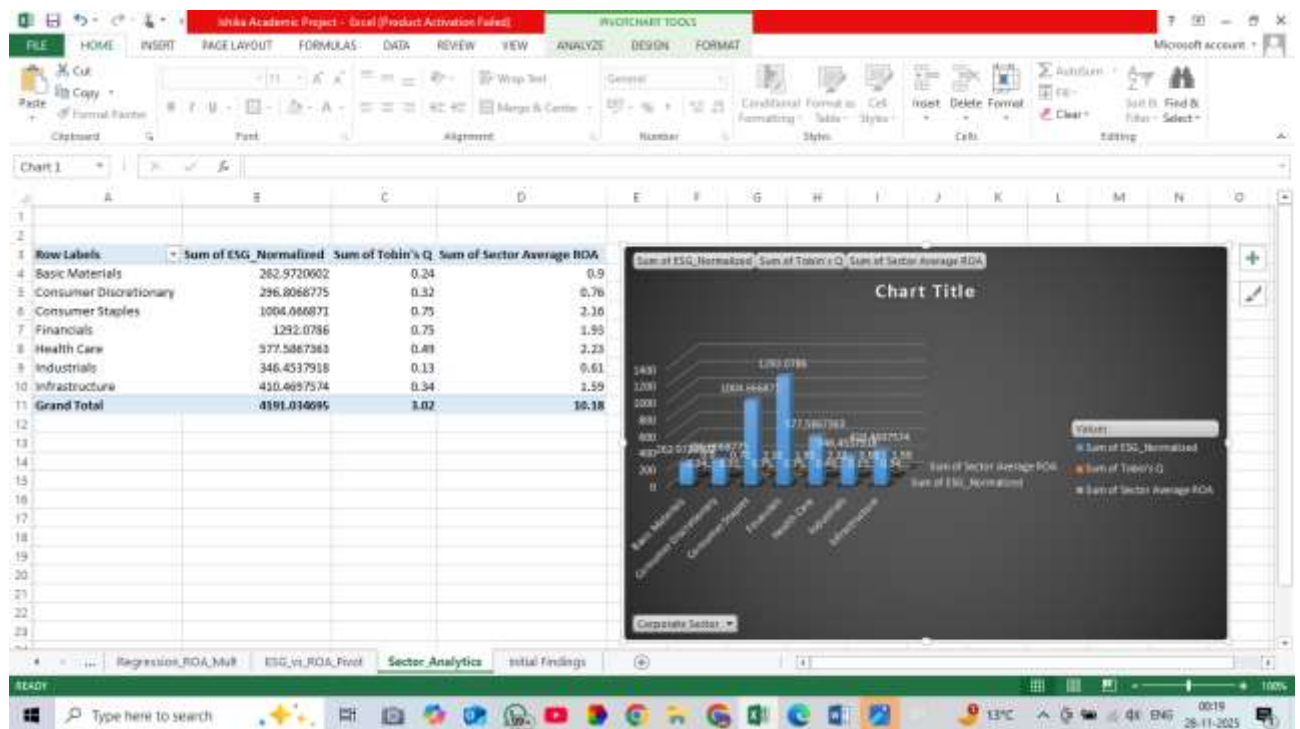
- Allows testing influence while controlling for financial structure and growth.

Quartile and Sector Analysis:

- ESG Quartile (Top, High, Medium, Low) mapped against financial KPIs.



- Sector-based comparisons of ESG maturity and financial outcomes.



Visualization and Dashboarding

Four dashboards were built using Tableau:

1) ESG Distribution Dashboard



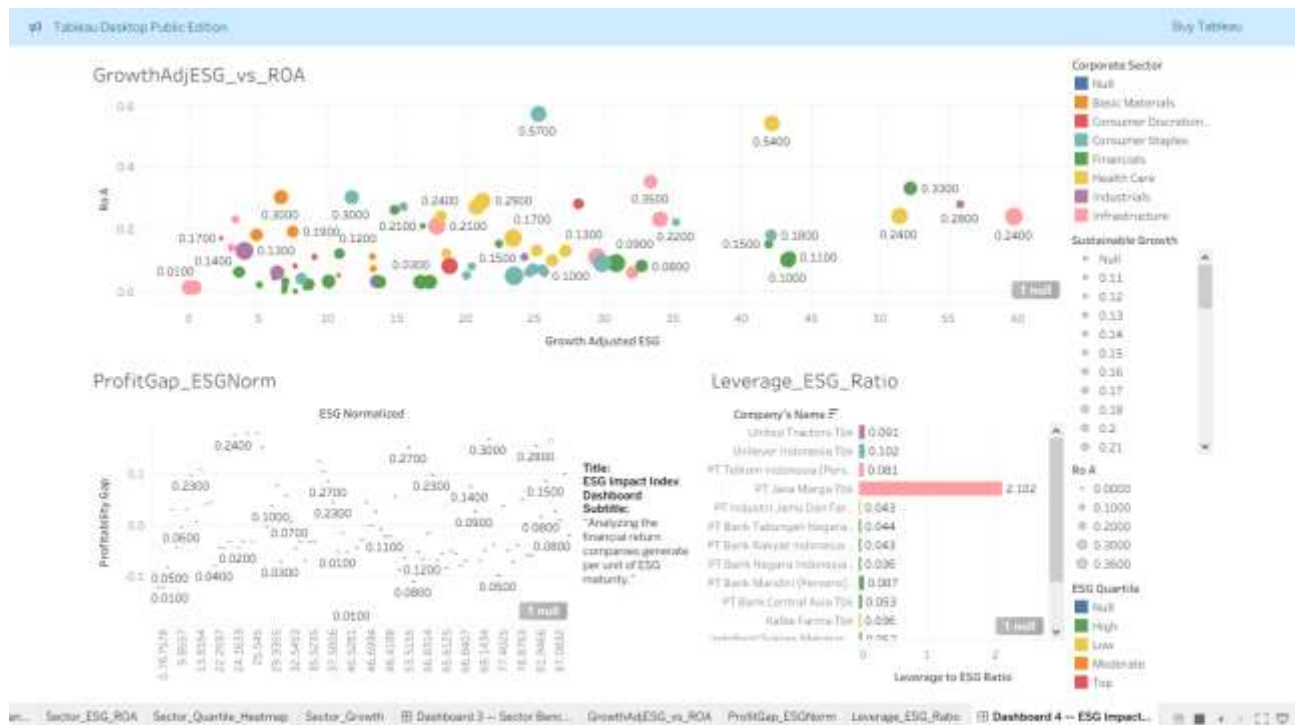
2) ESG vs Financial performance comparison



3) Sector Benchmarking Dashboard



4) ESG Impact Index Dashboard



These tools support real-time pattern identification, peer benchmarking, and decision-marking – simulating consulting deliverables.

Ethical Methodology Rigor

- All variables are operationally defined and measurable.
- No personal data is used; all sources are publicly available.
- Regression assumptions (linearity, homoscedasticity) were checked.
- Visual diagnostics complement statistical outputs for interpretability.